

Factors, dates, and strings

EVR 628- Intro to Environmental Data Science

Juan Carlos Villaseñor-Derbez (JC)

Table of contents

Exercise 0 - Set up	1
Exercise 1 - Dates	2
Exercise 2 - Strings	3
Part 1 - Data contained in a column with strings of text	3
Part 2 - Data STILL contained in a column with strings of text!	3
Exercise 3 - Visualization (a bit of everything)	4

Preamble:

Exercise 0 - Set up

Post-it up

1. Download this [CSV](#) file to your EVR 628 project folder. Save it into `data/raw/`.
2. Start a new script, call it `factors_dates_and_strings.R` and save it into your `scripts/01_processing/` folder
3. Add a code outline and load the `tidyverse` and `janitor` packages at the top of your script

Post-it down

```
## SET UP #####  
# Load packages -----  
library(janitor)  
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Exercise 1 - Dates

Post-it up

1. Load your data into an object called `tour_data` <- and clean the column names
2. Inspect your data. How many columns / rows? What are these columns?
3. Create a new object called `tour_data_clean` <- and add a new column called `date` that contains the date. We will build a large pipeline starting here.
4. Remove the `year`, `month`, and `day` columns that we no longer need
5. Place the `date` column all the way to the left

```
# Load data -----
tour_data <- read_csv("data/raw/tour_data.csv") |>
  clean_names()

dim(tour_data)
colnames(tour_data)
head(tour_data)

## PROCESSING #####
# Add a date -----
tour_data_clean <- tour_data |> #3) New object
  mutate(date = make_date(year, month, day)) |> # 4) Build my date
  select(date, everything(), -c(year, month, day)) # 5) Remove and organize columns
```

```
[1] 60 6
```

```
[1] "year"      "month"      "day"        "vessel"     "passengers"
[6] "notes"
```

```
# A tibble: 6 x 6
  year month   day vessel passengers notes
<dbl> <dbl> <dbl> <chr>      <dbl> <chr>
1  2025     1     1 Condor         21 8 Sea lions; 5 Whales; 4 Whale shark
```

2	2025	1	7 Falcon	6 5 whale; 7 Sea lions; 5 Sea turtle; 4 sea~
3	2025	1	13 Falcon	8 5 Dolphins
4	2025	1	19 Condor	6 4 Sea turtle; 4 sea turtle
5	2025	1	25 Condor	29 2 sea turtle; 8 whale
6	2025	2	1 Falcon	19 3 Whale shark; 7 whale

Post-it down

Exercise 2 - Strings

Part 1 - Data contained in a column with strings of text

Post-it up

1. As demonstrated in class, use `separate_longer_delim()` to create one row per species mentioned in the data
2. Standardize the data so all strings appear in lowercase
3. Make sure they are all singular

```
## PROCESSING #####
# Add a date -----
tour_data_clean <- tour_data |>
  mutate(date = make_date(year, month, day)) |>
  select(date, everything(), -c(year, month, day)) |>
  separate_longer_delim(cols = notes, delim = ";") |> # 1) Separate column into new rows
  mutate(notes = str_to_lower(notes),                # 2) All strings as lower case
         notes = str_remove(notes, "s$"))
```

Post-it down

Part 2 - Data STILL contained in a column with strings of text!

Post-it up

1. Look at the documentation for `str_extract()`
2. Run the first two examples at the bottom directly in your console. What seems to be going on?

pause to discuss

3. Using the `str_extract()` function, build a new column called `fauna` that contains the species observed¹
4. Print your data to the console. Does `fauna` look right?
5. Using the `str_extract()` function, build a new column called `n` that contains the number of organisms observed (make sure it's numeric)
6. Remove the `notes` column

```
## PROCESSING #####
# Add a date -----
tour_data_clean <- tour_data |>
  mutate(date = make_date(year, month, day)) |>
  select(date, everything(), -c(year, month, day)) |>
  separate_longer_delim(cols = notes, delim = ";") |>
  mutate(notes = str_to_lower(notes),
         notes = str_remove(notes, "s$"),
         fauna = str_extract(string = notes, pattern = "[a-z ]+$"), # 3) Extract fauna
         fauna = str_squish(fauna), # 4) Remove white space
         n = str_extract(string = notes, pattern = "[:digit:]+$"), # 5) Extract n
         n = as.numeric(n)) |> # and make sure it
  select(-notes) # 6) Remove notes
```

Post-it down

Exercise 3 - Visualization (a bit of everything)

Post-it up

1. Recreate the plot below

Post-it down

```
## VISUALIZE #####
ggplot(data = tour_data_clean,
       mapping = aes(x = str_to_sentence(fauna), # This is the only str-relevant change here
                     y = n)) +
  stat_summary(geom = "col", fun = "sum") +
  facet_wrap(~vessel, ncol = 1) +
  labs(title = "Sigthings by vessel",
       x = "Fauna",
       y = "N") +
  theme_minimal(base_size = 14)
```

¹Hint: Look at the documentation for `?regex`

Sightings by vessel

